SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0713

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

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Annexure A

Debugging & Optimisation Task

Code of Conduct:

I Harish Kumar / 192521340 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Debugging & Optimisation Task

Executive Summary

This assessment presents a full technical debugging and optimization analysis across five distinct enterprise networking scenarios, spanning WLAN/WWAN wireless infrastructure, Asynchronous Transfer Mode (ATM) backbones, Multiprotocol Label Switching (MPLS) networks, data-link error control in an industrial IoT setting, and end-to-end flow control for a large-scale online examination platform. Although the underlying technologies differ, a common engineering methodology is applied throughout each case study: (1) restate the operational scenario and the symptoms being reported, (2) perform root-cause analysis by mapping each symptom to an underlying technical deficiency, (3) outline the diagnostic/debugging approach a network engineer would follow to confirm those causes, and (4) recommend concrete, implementable optimization measures, supported by a before/after visual summary for each scenario.

Across all five cases, the recurring themes are: (a) capacity and load imbalance caused by static, non-adaptive configurations; (b) traffic that is not matched to the correct class-of-service or reliability mechanism for its nature (real-time vs. bulk, or error-sensitive vs. loss-tolerant); and (c) a lack of dynamic, feedback-driven control (load balancing, traffic engineering, flow control, or error correction) that would allow the network to adapt automatically to real-world operating conditions. The detailed analysis for each scenario follows below, concluding with a consolidated summary table and overall recommendations.

Question 1 — WLAN / WWAN Connectivity Optimization (Smart Office Building)

1.1 Scenario Recap

A six-storey smart office with 1,200+ employees relies on a WLAN for laptops, IP phones, printers and IoT devices, plus a 4G/5G WWAN for field engineers and sales executives working remotely. Reported symptoms include slow speeds at peak hours, frequent inter-floor Wi-Fi disconnections, weak signal in conference rooms, freezing video calls, uneven AP loading, unreliable WWAN in rural/highway areas, and devices still using outdated wireless security protocols.

1.2 Root Cause Analysis

- **Capacity/placement mismatch:** AP locations were likely planned without a proper RF site survey, so device density per floor does not match AP capacity — producing overloaded APs in some zones and idle ones elsewhere.
- **Coverage gaps:** Conference rooms typically have more walls, glass partitions and furniture that attenuate signal, and if APs were placed only for open-plan areas, these enclosed rooms fall into weak-signal zones.
- **Poor roaming behaviour:** Without fast BSS transition (802.11r) and client steering (802.11k/v), a laptop's radio 'sticks' to a weak, distant AP instead of a nearer, stronger one when the user changes floors, causing drops.
- **No traffic prioritization:** Video conferencing is latency- and jitter-sensitive; if QoS (WMM/DSCP) is not configured, video packets queue behind bulk downloads/uploads and the call freezes under load.

- **WWAN single-carrier dependency:** Field engineers likely rely on a single mobile carrier with no failover, so coverage gaps on highways/rural routes directly translate into total loss of connectivity.
- **Weak/legacy security configuration:** Some end devices are still associating using WEP or WPA(1)/WPA2-Personal instead of enterprise-grade WPA3/802.1X, exposing credentials and data to interception.

1.3 Diagnostic / Debugging Approach

- Conduct a full RF heat-map survey per floor to visualize signal strength, co-channel interference and dead zones.
- Pull AP association and utilization statistics from the Wireless LAN Controller (WLC) to quantify overloaded vs. underutilized APs.
- Review client roaming logs to measure hand-off latency and identify 'sticky client' patterns between floors.
- Inspect QoS/DSCP marking configuration on switches and APs to confirm whether real-time traffic is being prioritized.
- Analyze WWAN router logs and carrier signal-strength data along field engineers' regular travel routes.
- Audit the wireless security configuration (encryption type, authentication method) of every connected device via the WLC/NAC dashboard.

1.4 Optimization Recommendations

- Redesign AP placement using heat-map data; add APs in identified weak zones such as conference rooms, and enable automatic RF Resource Management (RRM) for dynamic power/channel tuning and load balancing.
- Enable 802.11r/k/v for fast, seamless roaming so devices hand off to the best AP automatically as employees move between floors.
- Apply WMM/DSCP-based QoS policies to prioritize VoIP and video conferencing traffic ahead of bulk data transfers.
- Deploy dual-SIM / multi-carrier SD-WAN routers for field engineers with automatic failover between carriers, supplemented by signal boosters on known weak routes.
- Enforce WPA3-Enterprise with 802.1X/RADIUS authentication across all devices, and retire support for legacy WEP/WPA on the network.
- Centralize monitoring through the WLC for ongoing visibility into AP load, client experience and security posture.

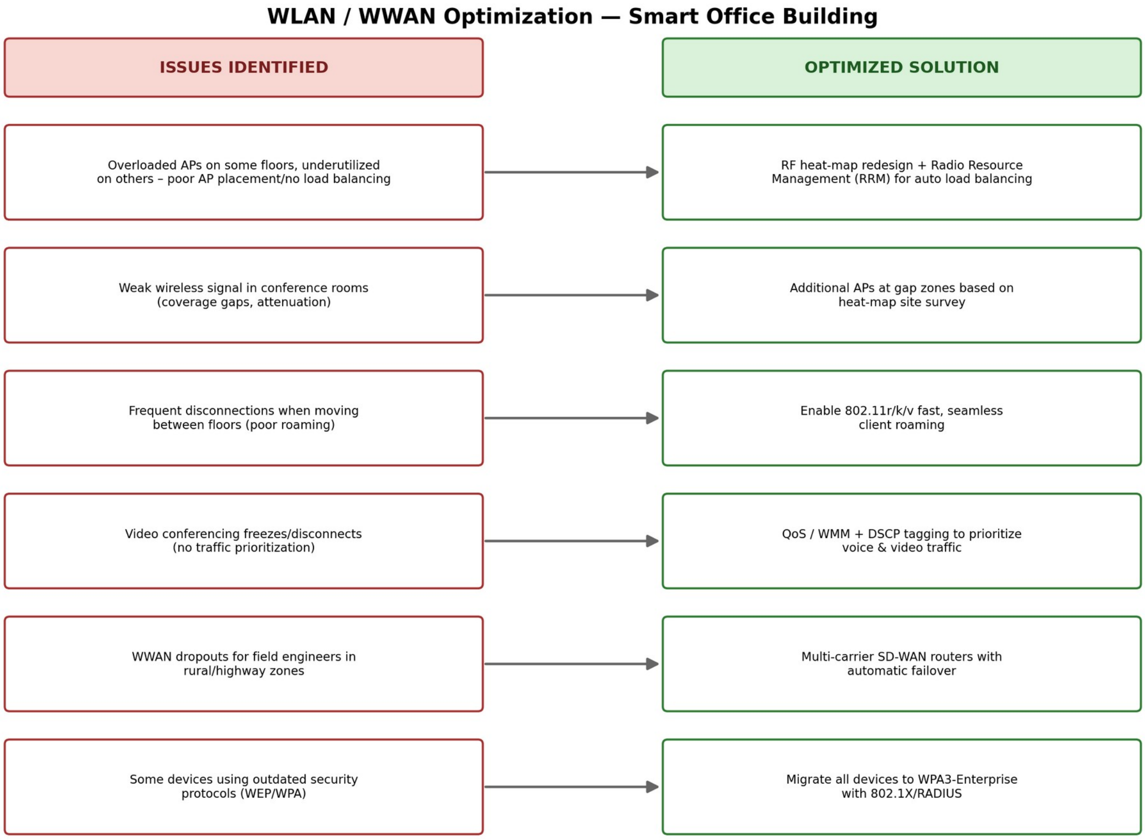

Figure 1: Identified WLAN/WWAN issues mapped to their corresponding optimization measures.

1.5 Expected Outcome

After implementation, AP load should be evenly distributed across floors, conference rooms should achieve full signal coverage, roaming-related disconnections should drop sharply, video conferencing should remain stable under peak load, field engineers should retain reliable connectivity through carrier failover, and all wireless traffic should be secured under WPA3-Enterprise.

Question 2 — ATM Network Debugging (Financial Enterprise)

2.1 Scenario Recap

A multinational financial organization runs online banking, VoIP and video conferencing over an ATM network using fixed 53-byte cells and Virtual Circuits (VCs) for QoS. During peak hours, transactions are delayed, voice calls suffer jitter and packet loss, video quality drops, several switches run at high utilization, some VCs are congested, and bandwidth is not efficiently allocated across traffic classes.

2.2 Root Cause Analysis

- **VC congestion at peak load:** VC bandwidth (Peak/Sustainable Cell Rate) was likely provisioned for average, not peak, demand, so cells queue and get delayed or dropped when traffic spikes.

- **Incorrect ATM service category:** Real-time voice/video appears to be riding on UBR (Unspecified Bit Rate, best-effort) instead of CBR or real-time VBR, which offer no delay/jitter guarantees — explaining the reported jitter and packet loss.
- **Uneven switch utilization:** Static VC/VP routing does not redistribute load, so some switches are saturated while others sit idle.
- **Lack of admission control:** Without traffic shaping or call admission control at the UNI, VCs can be oversubscribed beyond the physical link's real capacity.

2.3 Diagnostic / Debugging Approach

- Collect per-VC ATM performance metrics — Cell Loss Ratio (CLR), Cell Transfer Delay (CTD) and Cell Delay Variation (CDV) — via the switch's ATM MIB/SNMP interface.
- Cross-reference each application (banking, VoIP, video) against its assigned ATM service category (CBR, rt-VBR, nrt-VBR, ABR, UBR) to spot mismatches.
- Review switch utilization reports to identify consistently congested vs. underused switches.
- Trace VC/VP provisioning records to compare allocated PCR/SCR bandwidth against actual measured demand.

2.4 Optimization Recommendations

- Reassign VoIP and video conferencing to CBR or real-time VBR service categories with guaranteed bandwidth and bounded delay/jitter; keep bulk/non-critical transfers on UBR/ABR.
- Increase PCR/SCR allocation on congested VCs, and provision additional VCs across less-utilized switches to spread the load.
- Implement traffic shaping and call admission control at the UNI to prevent oversubscription of any single VC or switch.
- Introduce dynamic VC/VP re-routing so traffic automatically shifts away from congested switches during peak periods.
- Establish ongoing CLR/CDV monitoring with alert thresholds so QoS degradation is caught before it affects customers.

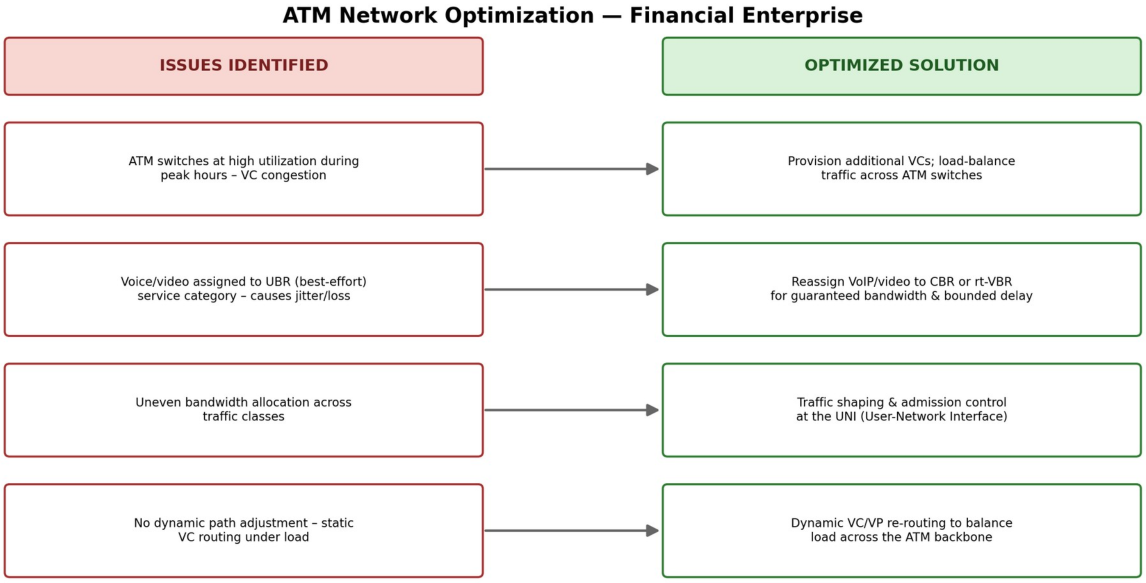

Figure 2: ATM network issues mapped to their corresponding optimization measures.

2.5 Expected Outcome

With correctly assigned service categories and better-balanced VC provisioning, voice and video should regain low jitter and near-zero cell loss, online banking transactions should process without delay, and overall switch utilization should be more evenly distributed across the ATM backbone.

Question 3 — MPLS Network Debugging (E-Commerce Backbone)

3.1 Scenario Recap

An e-commerce company connects its headquarters, regional offices, warehouses and cloud data centers over MPLS, carrying order processing, ERP, video conferencing, VoIP, inventory management and customer support traffic. During peak hours, cloud applications are slow, orders are delayed, voice quality drops and video conferences are interrupted. Monitoring shows some Label Switched Paths (LSPs) heavily congested while others are underused, incorrect label mappings on some routers, and no traffic engineering configured.

3.2 Root Cause Analysis

- **Static/manual LSP path selection:** Without MPLS Traffic Engineering (MPLS-TE), traffic naturally funnels onto the shortest/default LSPs regardless of their current load, leaving parallel paths idle.
- **Incorrect label mappings:** Misconfigured LDP/RSVP-TE bindings on certain routers can cause packets to be forwarded along suboptimal or looping paths, inflating latency.
- **No constraint-based routing:** Without CSPF (Constrained Shortest Path First) and bandwidth reservation via RSVP-TE, LSPs are not established with real available-bandwidth awareness.

- **No fast failure/congestion recovery:** Absence of Fast Reroute (FRR) means congestion or link issues are not automatically routed around, prolonging the impact of any single congested link.

3.3 Diagnostic / Debugging Approach

- Inspect each router's Label Forwarding Information Base (LFIB) and LDP bindings to confirm they match the intended label distribution.
- Use MPLS ping/traceroute along suspect LSPs to detect misrouted or looping label-switched paths.
- Query the Traffic Engineering Database (TED) to compare reserved vs. available bandwidth per LSP and identify chronically congested paths.
- Review LDP/RSVP-TE session configuration consistency across all routers in the MPLS domain.

3.4 Optimization Recommendations

- Correct the faulty label bindings identified on affected routers and re-verify LFIB consistency network-wide.
- Enable MPLS-TE with CSPF so LSPs are computed based on real-time available bandwidth, automatically balancing load across the backbone.
- Establish multiple load-balanced LSPs between high-traffic site pairs (HQ ↔ data centers ↔ regional offices) instead of relying on a single default path.
- Enable MPLS Fast Reroute (FRR) to provide sub-50ms failover around congested or failed links.
- Deploy ongoing MPLS OAM monitoring (latency, packet loss per LSP) to catch emerging congestion before customers are affected.

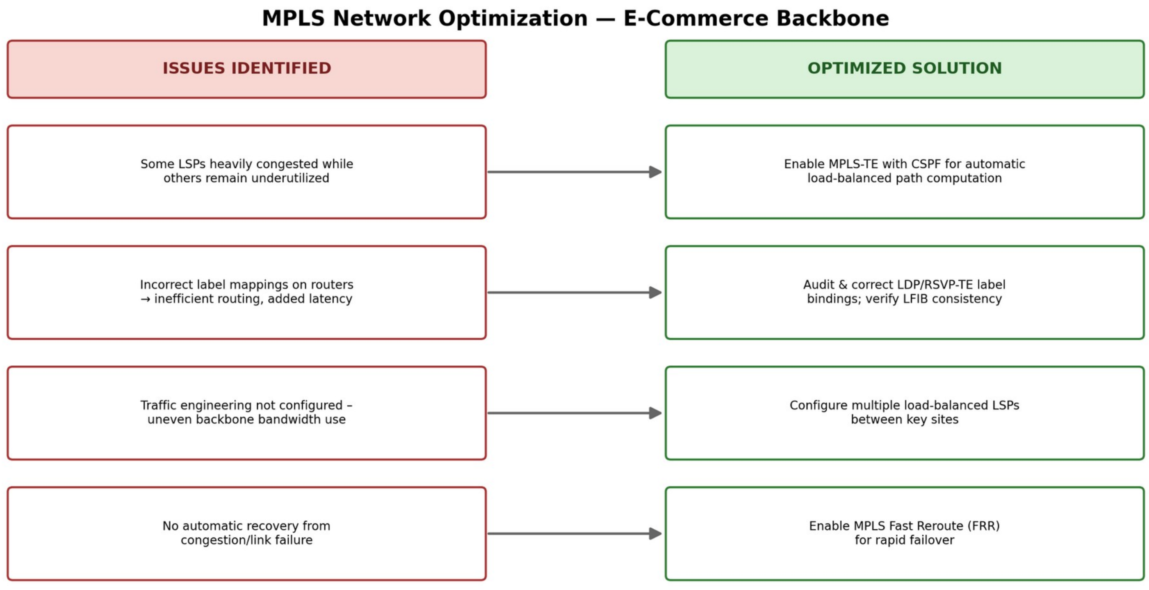

Figure 3: MPLS network issues mapped to their corresponding optimization measures.

3.5 Expected Outcome

Correcting label mappings and enabling traffic-engineered, load-balanced LSPs should even out backbone utilization, restore fast access to cloud applications and ERP systems, and stabilize voice and video conferencing quality even during peak business hours.

Question 4 — Error Control Optimization (Automated Warehouse Network)

4.1 Scenario Recap

In an automated warehouse, barcode scanners, RFID readers, autonomous robots and inventory systems continuously exchange shipment and stock data with a central server. Some records arrive with incorrect information, barcode scans occasionally fail, and inventory counts mismatch the physical stock. Monitoring during peak hours shows increased transmission errors from electrical interference, signal attenuation and congestion; some corrupted frames are caught and retransmitted, but others are not, causing bad data to reach the database.

4.2 Root Cause Analysis

- **Weak error-detection code:** A simple parity bit or basic checksum has a low Hamming distance and cannot reliably catch burst errors — common in electrically noisy warehouse environments — letting some corrupted frames pass as 'clean'.
- **No forward error correction:** The current scheme can only detect (not correct) errors, so every genuinely caught error requires a retransmission round-trip, adding delay under high scan volume.
- **Physical-layer noise sources:** Motors, conveyor systems and RF-heavy equipment introduce electromagnetic interference (EMI) and attenuation, directly raising the raw bit-error rate (BER) on both wired and wireless links.
- **Inconsistent retransmission handling:** If the ARQ (Automatic Repeat reQuest) mechanism is not implemented consistently (e.g., simple stop-and-wait without proper sequencing), some detected errors may still not trigger a correct retransmission.

4.3 Diagnostic / Debugging Approach

- Compare CRC/parity failure counts in network logs against actual downstream data-mismatch reports to quantify how many errors are slipping through undetected.
- Run controlled interference tests (e.g., near active machinery vs. away from it) to correlate bit-error rate with EMI sources.
- Mathematically evaluate the current error-detection code's Hamming distance to confirm its weakness against burst/multi-bit errors.
- Trace the ARQ behaviour end-to-end (via packet capture) to confirm whether every detected error actually results in a retransmission.

4.4 Optimization Recommendations

- Replace the existing parity/checksum scheme with CRC-32, which offers far stronger burst-error detection suited to this noisy environment.

- Introduce Hamming or Reed-Solomon forward error correction (FEC) on the more error-prone RFID/barcode links, allowing many errors to be corrected without a retransmission round-trip.
- Implement a reliable Selective-Repeat ARQ scheme so only genuinely corrupted frames are retransmitted, minimizing redundant traffic and delay.
- Physically shield or reroute cabling away from high-EMI machinery, use twisted-pair/shielded cable, and select a cleaner RF channel for wireless scanners.
- Add an application/database-level checksum as a secondary safety net to catch any residual error that slips past the link layer.

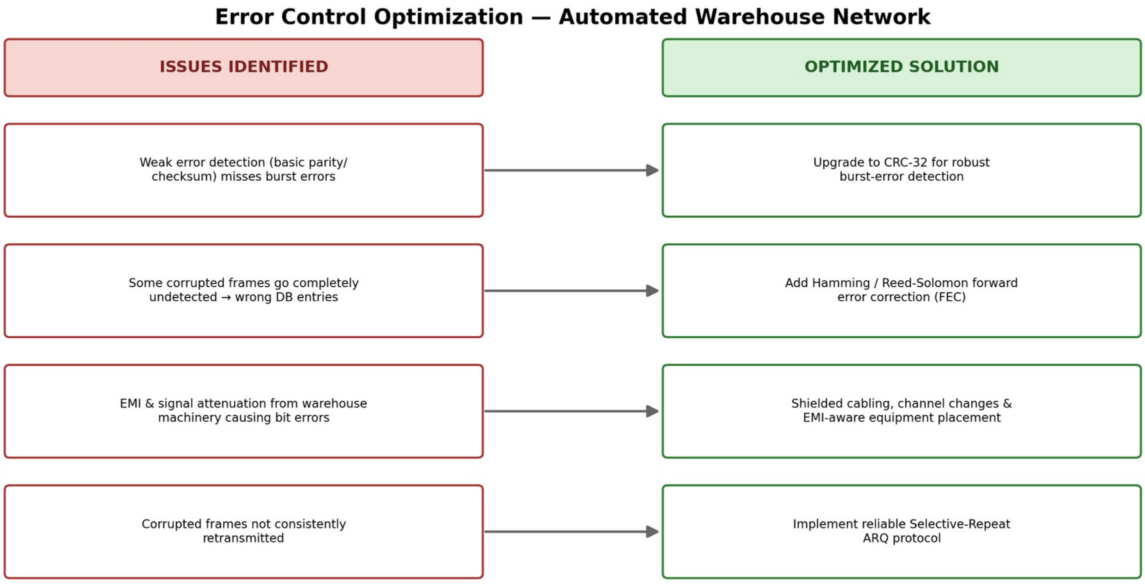

Figure 4: Error-control weaknesses mapped to their corresponding optimization measures.

4.5 Expected Outcome

With stronger detection (CRC-32), added correction (FEC) and a reliable ARQ scheme, undetected errors should fall close to zero, inventory records should match physical stock far more consistently, and barcode-scan failures and processing delays should be significantly reduced.

Question 5 — Flow Control Optimization (Online Examination Platform)

5.1 Scenario Recap

An online education platform serves 15,000+ students during live exams, with a central server on a 10 Gbps link sending question papers, images, video and acknowledgements to students on much slower and varied connections. During peak load, students experience incomplete downloads, delayed loading, duplicate packets, repeated retransmissions, receiver buffer overflow and application crashes; administrators observe excessive retransmissions, higher latency, lower throughput and poor bandwidth utilization.

5.2 Root Cause Analysis

- **Sender-receiver speed mismatch:** The server transmits at a rate suited to its own 10 Gbps capacity without adapting to each student's actual receive capability, overwhelming slower clients.
- **Ineffective flow control:** The sliding-window mechanism (if present) does not appear to be honoring each receiver's advertised window size, so the server keeps pushing data faster than clients can drain their buffers.
- **Receiver buffer overflow:** Once a client's buffer fills, further incoming segments are dropped, which is consistent with the observed application crashes and incomplete downloads.
- **Retransmission storm:** Buffer-overflow packet loss triggers retransmissions; if the retransmission timeout (RTO) is too aggressive or congestion control is weak/absent, this creates duplicate packets and even more traffic, worsening the very congestion that caused the problem.
- **Single-origin bottleneck:** All 15,000 students are served from one central server/location, so even efficient flow control cannot fully offset the combined bandwidth-delay product of a globally distributed, simultaneous audience.

5.3 Diagnostic / Debugging Approach

- Capture packet traces (e.g., with Wireshark) during a peak exam window to measure retransmission rate, duplicate ACKs and round-trip time (RTT).
- Check whether the server's transport stack correctly honors each client's advertised receive window (rwnd), or is sending irrespective of it.
- Profile student connection types (fiber, Wi-Fi, mobile) and correlate performance issues with each bandwidth tier.
- Determine whether transmission is using TCP with proper congestion control, or a less-adaptive transport (e.g., unmanaged UDP) that bypasses flow/congestion control entirely.

5.4 Optimization Recommendations

- Enforce proper sliding-window flow control so the server dynamically throttles its transmission rate to each client's advertised window, preventing buffer overflow.
- Deliver content adaptively/in chunks (similar to adaptive-bitrate streaming) so slower clients receive smaller, more manageable segments instead of one large burst.
- Apply server-side rate limiting or traffic shaping per client group, based on detected connection type.
- Distribute load via a Content Delivery Network (CDN) or regional edge servers so no single 10 Gbps origin server has to serve all 15,000 students directly.
- Tune retransmission timeout (RTO) and congestion-control behaviour, or migrate to a protocol such as QUIC/HTTP-3 that handles variable-bandwidth clients and packet loss more gracefully.
- Stagger content delivery (e.g., releasing question papers in smaller sequential chunks with slight randomized offsets) to avoid one simultaneous burst to all students.

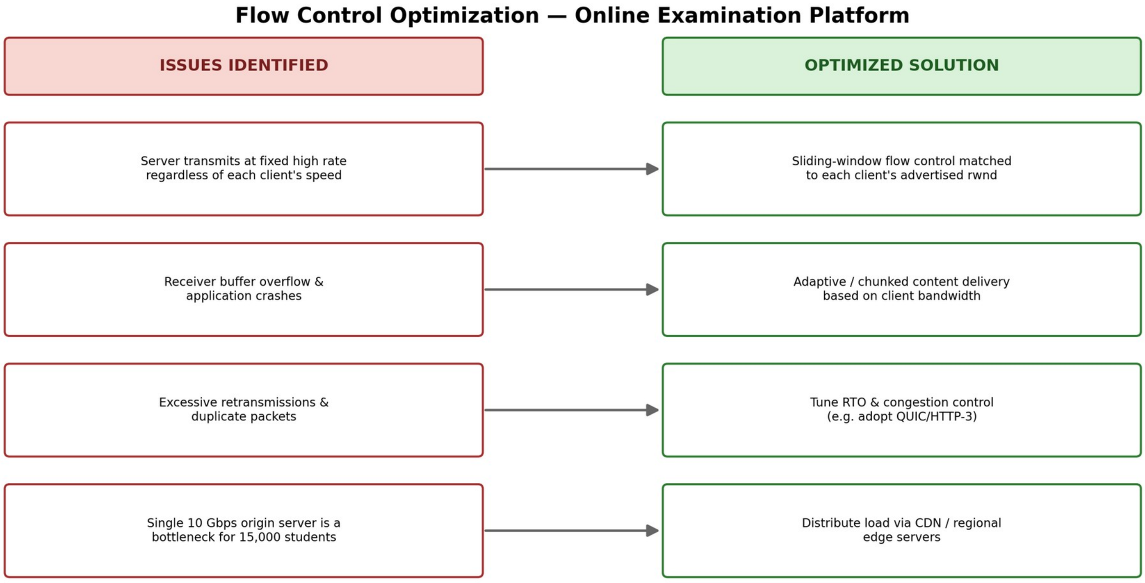

Figure 5: Flow-control issues mapped to their corresponding optimization measures.

5.5 Expected Outcome

With rate-adaptive, window-based flow control and distributed delivery, receiver buffer overflow and application crashes should be eliminated, retransmissions and duplicate packets should drop sharply, and all students — regardless of connection speed — should receive exam content reliably and on time.

Consolidated Summary of Findings

The table below consolidates the primary technology, main reported issue, underlying root cause and key recommended fix for each of the five scenarios analyzed in this assessment.

Q	Technology	Primary Issue	Root Cause	Key Recommended Fix
1	WLAN / WWAN	Disconnections, weak signal, freezing calls	Poor AP placement, no fast roaming/QoS, single-carrier WWAN	Heat-map redesign, 802.11r/k/v, QoS, multi-carrier SD-WAN, WPA3
2	ATM Network	Transaction delay, voice jitter, poor video	Wrong service category (UBR for real-time), VC congestion	Reassign to CBR/rt-VBR, admission control, load-balanced VCs
3	MPLS Network	Slow cloud apps, delayed orders, call issues	No traffic engineering, incorrect label mappings	Enable MPLS-TE/CSPF, fix LDP bindings, enable FRR
4	Warehouse LAN (Error Control)	Wrong data, failed scans, stock	Weak error detection, no FEC,	CRC-32 + Hamming/Reed-

		mismatch	EMI/attenuation	Solomon FEC, shielded cabling, Selective-Repeat ARQ
5	Exam Platform (Flow Control)	Buffer overflow, retransmissions, crashes	No adaptive flow control, single-origin bottleneck	Window-based flow control, adaptive delivery, CDN distribution

Overall Conclusion

Although these five scenarios involve different network technologies — wireless access, ATM, MPLS, industrial data-link communication, and application-layer flow control — they share a common engineering lesson: reliable, high-performance networks require configurations that adapt dynamically to real operating conditions rather than relying on static, one-size-fits-all setups. In every case, the recommended fixes revolve around three principles that generalize well beyond these specific scenarios: (1) match each traffic type to the control mechanism it actually needs (QoS class, error-correction strength, or flow-control aggressiveness), (2) monitor continuously so load imbalances and misconfigurations are caught early, and (3) build in adaptive/redundant capacity (load balancing, traffic engineering, FEC, or distributed delivery) so the network degrades gracefully instead of failing under peak demand.